

DIVYANSHU JAIN

PhD Candidate specializing in Energy Economics, causal inference, machine learning and experimentation
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EDUCATION

University of California, Davis

PhD, Economics

Sep 2022 - Apr 2027 (Expected)

Delhi School of Economics

M.A., Economics

Aug 2018 - Aug 2020

BITS Pilani

M.Sc. (Hons.), Economics

Jul 2011 - Apr 2015

RESEARCH PROJECTS

Storing the Sun: The Impacts of Renewable Energy on Wholesale Prices in the Era of Storage

- Investigating changes in causal impacts of daily solar generation due to BESS expansion on wholesale electricity prices and generator revenues in CAISO.
- Showed that 1 GW of increase in daily solar generation leads to decline of \$0.4 across hours in wholesale prices in CAISO in 2021-2024, no longer generating evening peaks.

Environmental Justice Effects of RGGI: Source-Level Emissions, Spillovers, and Exposure Inequality, with Xinghuan Luo

- Estimate RGGI impacts on facility-level CO₂, SO₂, and NO_x emissions using difference-in-differences and PPML counterfactuals, recovering source-specific abatement at regulated plants.
- Measure spillovers to nearby non-RGGI PJM facilities and link emissions changes to demographic exposure data to assess whether market-based climate policy shifted pollution burdens.

Health Impacts of Coal Plant Retirements in the United States, with Xinghuan Luo

- Link coal plant retirement and emissions data with restricted-use mortality records and InMAP source-receptor matrices to measure exposure based on pollution transport.
- Use exposure-based two-stage difference-in-differences with spillover adjustments across active and retired plants to estimate public health gains from coal retirements.

Multi-Agent Reinforcement Learning for Electricity Markets

- Developed a multi-agent RL framework to study tacit collusion in electricity-market duopolies, where independent firms learn pricing strategies without centralized coordination.
- Benchmarked learned pricing against competitive, Cournot, SFNE, and monopoly outcomes; designed a Collusion Index and counterfactual exercises showing how discounting, exploration, and cost asymmetry shape supra-competitive pricing.

Economic Uncertainty Index from WSJ Text

- Engineered an end-to-end NLP pipeline on WSJ articles using LLM-based semantic topic extraction.
- Constructed a daily uncertainty index by computing cosine distance between consecutive sparse topic vectors and day-over-day topic-count drift, quantifying shifts in macroeconomic attention over time.

EXPERIENCE

Research Assistant, UC Davis (Experiment on EV Charging Behavior)

April 2025 - May 2026

- Built household-level panel datasets from telemetry and utility usage records linked to randomized treatment assignments.
- Estimated treatment effects using A/B testing methods, Difference-in-Differences, IV, and heterogeneous treatment effect models for uplift measurement.

Consultant, World Bank

Mar 2022 - Aug 2022

- Built cross-country panel datasets on credit markets and macro outcomes across emerging economies.
- Estimated dynamic effects of credit supply shocks using local projections with fixed effects.

Associate Product Manager, Mande (Agri E-commerce)

Jul 2017 - Jul 2018

- Led growth and operations initiatives in a two-sided marketplace; partnered with business and engineering teams.

Program Manager, Belong.co (HR Tech)

May 2015 - Apr 2017

- Managed cross-functional product operations and analytics initiatives for hiring marketplace workflows.

TECHNICAL SKILLS

Languages: SQL, Python, R, MATLAB

Libraries: pandas, NumPy, scikit-learn, statsmodels, PyTorch, XGBoost, matplotlib

Methods: Experimentation, Causal Inference, DiD, IV/2SLS, Forecasting, Reinforcement Learning, NLP

Tools: Git, Jupyter, Stata, L^AT_EX